

0 clubs; 4 hours.

Problem 1 (OMMC Feb. 2022)

Plugging in $x = y = 0$, we get $f(0) = 0$.

Plugging in $x = 1$, we get $f(1 - y) + f(y) = 1$. Therefore, $f(1) = 1$.

We now plug in $x = y + 1$ to get,

$$\begin{aligned} f(1) + (y + 1)f(y) + f(y) &= (y + 1)^2 \\ \Leftrightarrow (y + 2)f(y) &= y(y + 2) \end{aligned}$$

Therefore, for all $y \neq -2$, $f(y) = y$.

Plugging in $x = 0$, we get $-f(-y) = f(y) \Leftrightarrow f(-2) = -f(2) = -2$. Therefore, $f(x) \equiv x$.

Problem 2 (Napkin 81A)

By Cauchy's equation of Q , we have, for any positive p, q ,

$$f\left(\frac{p}{q}\right) = \frac{p}{q}$$

Therefore,

$$\begin{aligned} f\left(\frac{22}{7}\right) &= \frac{22}{7} \\ \Leftrightarrow f\left(\frac{22}{7} - \pi\right) &= 0 \\ \Leftrightarrow f\left(790.8\left(\frac{22}{7} - \pi\right)\right) &= 0 \\ \Leftrightarrow f(1) - f\left(790.8\left(\frac{22}{7} - \pi\right)\right) &= 1 \\ \Leftrightarrow f\left(\frac{3954\pi}{5} - \frac{86953}{35}\right) &= 1 \end{aligned}$$

Since the left-hand-side is less than 10^{-5} ,

$$\Leftrightarrow f\left(9000\left(\frac{3954\pi}{5} - \frac{86953}{35}\right)\right) = 9000$$

The LHS here is less than 1. We are done.

Problem 3 (Napkin 81B)

Weak "proofs" more like intuition. The answer for all of these is yes.

1. Seems intuitive since we can "copy paste" ranges by shifting.
2. We can have $x = 0$, and $y > 0$ to get $f(x) = f(0) + f(x) \Leftrightarrow f(0) = 0$.
Therefore, $f(0) = f(x) + f(-x) \Leftrightarrow f(x) = -f(-x)$.
3. f is bounded below at 0.
4. Let $f(1) = x$, f would be bounded below at x .
5. If $f(x) = 0$ for any x , then $f(y) = f(x + y)$ so not injective. If not, then not surjective.
6. Have $f(x) = 0$ for no x , quite possible.
7. Should be possible since we can get very small amounts by subtracting if f is not linear.
8. Rational + rational = rational so should be possible.
9. Have the small values be irrational so they stack up to form bigger irrationals.

13 clubs; 5 hours.

Problem 4

We claim the answer is either $f \equiv 0$ or $f(x) = x^3$. Both of these work.

We plug in $y = 0$ to get,

$$0 = 3x f(x) f(0)$$

Case 1: $x f(x) = 0$

This means that $f(x) = 0$ for all real $x \neq 0$. We plug in $x = 1$; $y = -1$ to get,

$$(f(0) - f(1) - f(-1)) = 0 \Leftrightarrow f(0) = 0$$

Therefore, $f \equiv 0$ in this case.

Case 2: $f(0) = 0$

We plug in $y = -x$.

$$\begin{aligned} x^4(f(0) - f(x) - f(-x)) &= 0 \\ \Leftrightarrow x^4(f(x) + f(-x)) &= 0 \end{aligned}$$

Therefore, f is odd.

Let $f(1) = k$. We plug in $y = 1$ to get,

$$x^2(f(x+1) - f(x) - k) = 3k(x+1)f(x)$$

We next plug in $y = -x - 1$ to get,

$$x^2(x+1)^2(-k-f(x)+f(x+1)) = -3f(x)f(x+1)$$

We divide the second equation by the first.

$$-(x+1)^2 = \frac{f(x+1)}{-k(x+1)} \text{ for } x \neq 0$$

$$\Leftrightarrow f(x+1) = k(x+1)^3 \text{ for } x \neq 0, f(1) = k$$

$$\Leftrightarrow f(x+1) = k(x+1)^3$$

Therefore, $f(x) = kx^3$. If we plug this in, we will simplify to get,

$$3k(x^4y^3 + x^3y^4) = 3k^2(x^4y^3 + x^3y^4)$$

If $k = 0$, then $f \equiv 0$. Otherwise, $k = 1$ and $f(x) = x^3$.

These are the only solutions.

Problem 5

We claim that $f(x) = x^2 + kx$ for some real k .

All such solutions work since $x^2 + 2xy + y^2 + kx + ky = (x+y)^2 + k(x+y)$.

We let $g(x) := f(x) - x^2$. Simplifying,

$$f(x) + f(y) + 2xy = f(x+y)$$

$$\Leftrightarrow g(x) + g(y) + x^2 + 2xy + y^2 = g(x+y) + (x+y)^2$$

$$\Leftrightarrow g(x) + g(y) = g(x+y)$$

Since g is bounded above -1 on $(0, 1)$, $g(x) = kx$ for some k .

Therefore, $f(x) = x^2 + kx$ for some k .

Problem 6 (IMO 2019/1)

We claim that the solutions are $f \equiv 0$ and $f(x) = 2x + c$ for some c , which obviously work.

Plugging in $a = 0; b = x + y$, we get,

$$f(0) + 2f(x+y) = f(f(x+y))$$

Therefore,

$$f(2a) + 2f(b) = f(f(a+b))$$

$$\Leftrightarrow f(0) + 2f(a + b) = f(2a) + 2f(b)$$

We plug in $a = 1$ to get,

$$f(0) - f(2) = 2f(b) - 2f(b + 1)$$

Since the LHS is constant, we get that f is linear.

We express $f(x) = mx + k$ to get, and plug this into,

$$f(0) + 2f(b) = f(f(b))$$

$$\Leftrightarrow 2mb + 3k = bm^2 + mk + k$$

$$\Leftrightarrow 2mb + 2k = bm^2 + mk$$

Looking at the coefficient for b , $2m = m^2$. Therefore, this is only always true when either $m = 2$, or $m = k = 0$.

20 clubs; 6.5 hours.

Problem 7 (Iran TST 1996)

We claim that our solutions are $f(x) = 0$ and $f(x) = x^2$, both of which work.

We let $y = \frac{x^2 - f(x)}{2}$ to get,

$$0 = 2f(x)(x^2 - f(x))$$

$f(x) \in \{0, x^2\}$ for all x . We prove that either $f(x) = 0$ or $f(x) = x^2$.

Case 1: $f(1) = 1^2 = 1$

We plug in $x = 1$ to get,

$$f(y + 1) = f(1 - y) + 4y$$

For $y \neq 0, -1$, the RHS is non-zero, so $f(x) = x^2$ for $x \neq 0$. However, at 0, $0 = x^2$, so $f(x) \equiv x^2$.

Case 2: $f(1) = 0$

$$f(y + 1) = f(-y)$$

If the right hand-side is non-zero for any y , then $f(y + 1) = y^2$. Since $f(y + 1) > 0$, $f(y + 1) = (y + 1)^2$. Therefore, $(y + 1)^2 = y^2$, which is impossible for $y \neq -\frac{1}{2}$.

Knowing that $f(x) = 0$ for $x \neq -\frac{1}{2}$, we plug in $x = \frac{1}{2}; y = -\frac{1}{4}$ to get $f\left(-\frac{1}{2}\right) = 0$. Therefore, in this case, $f \equiv 0$.

Problem 8 (MOP 2017)

We claim that $f(x) = x + c$ for some c , which obviously works.

We plug in $y = 0$ to get $f(f(x)) = f(x) + f(0)$. Therefore,

$$f(f(x+y)) = f(x+y) + f(0) = f(x) + f(y)$$

We let $g(x) = f(x) - f(0)$. Therefore,

$$g(x+y) = g(x) + g(y)$$

Since g is strictly increasing, $g(x) = kx$ for some k by Cauchy's equations.

We let $c = f(0)$ to get $f(x) = kx + c$. Plugging this into the original equation,

$$k^2(x+y) + ck + c = kx + ky + 2c$$

We have either $k = 0$ or $k = 1$. However, f is strictly increasing, so $k = 1$. Therefore, $f(x) = x + c$.

Problem 9 (Russia 2000/11.1)

We claim that the only solution is $f \equiv c$ for some c .

We plug in $y = z = 0; z = a$ to get,

$$3f(a) \leq 2f(a) + f(0) \Leftrightarrow f(a) \leq f(0)$$

We plug in $x = y = \frac{a}{2}; z = -\frac{a}{2}$ to get,

$$3f(0) \leq f(a) + 2f(0) \Leftrightarrow f(0) \leq f(a)$$

Therefore, $f(a) = f(0)$. We let $c = f(0)$ to be done.

28 clubs; 8 hours.

Problem 10 (ELMO SL 2010 A3)

We claim the only solution is $f(x) = x$, which obviously works as $\min(x, y) + \max(x, y) = x + y$.

Note that,

$$\begin{aligned} f(x+y) &= \max(f(x), y) + \min(f(y), x) \\ \Leftrightarrow f(y+x) &= \max(f(y), x) + \min(f(x), y) \end{aligned}$$

Adding these two, since $\min(x, y) + \max(x, y) = x + y$,

$$2f(x+y) = f(x) + f(y) + x + y$$

We let $g(x) := f(x) - x$,

$$2g(x+y) = g(x) + g(y)$$

We plug in $y = 0$ to get $g(x) = g(0)$, so $g(x) = c$.

Therefore, $f(x) = x + c$ for some c . We prove that $c = 0$.

Plugging back into original, we get,

$$x + y + c = \max(x + c, y) + \min(x, y + c)$$

For any $c > 0$, this is impossible when we plug in $x = 0, y = 2c$ since $0 + 2c + c \neq 2c + 0$.

For any $c < 0$, this is also impossible when we plug in $x = 0, y = -2c$ since $-c \neq -2c + 0$.

Problem 11 (IMO 2008/4)

We claim the only solutions are $f(x) = \frac{1}{x}$ and $f(x) = x$. Since $wx = yz \Leftrightarrow \frac{w^2 + x^2}{y^2 + z^2} = \frac{\frac{1}{x^2} + \frac{1}{w^2}}{\frac{1}{y^2} + \frac{1}{z^2}}$, these both works.

Plugging in all 1's, $f(1) = 1$.

We plug in $w = 1; x = t; y = z = \sqrt{t}$ to get,

$$\begin{aligned}\frac{1 + f(t)^2}{2f(t)} &= \frac{1 + t^2}{2t} \\ \Leftrightarrow f(t) + t^2 f(t) &= t + t f(t)^2 \\ \Leftrightarrow 0 &= t \left(f(t) - \frac{1}{t} \right) (f(t) - t)\end{aligned}$$

Since $t > 0$, either $f(t) = \frac{1}{t}$ or $f(t) = t$.

Problem 12 (JMO 2019/2)

We claim the only solutions are if $a = \pm b$.

We can achieve $a = b$ when $f(x) = g(x) = x + \frac{a}{2}$.

We can achieve $a = -b$ when $f(x) = -x + \frac{a}{2}$ and $g(x) = -x - \frac{a}{2}$.

Claim: f, g are bijective

Proof:

f, g are obviously surjective since to get anything we just plug in $f(g(x - a))$.

Since g takes all integers, we can prove injectivity by proving,

$$f(g(x_0)) = f(g(x_1)) \Leftrightarrow g(x_0) = g(x_1)$$

We can do this as follows:

$$f(g(x_0)) = f(g(x_1)) \Leftrightarrow x_0 = x_1 \Leftrightarrow g(x_0) = g(x_1)$$

We now continue with the problem.

$$f(g(f(x))) = f(x + b) = f(x) + a$$

We independently prove that neither of $|a| > |b|$ and $|a| < |b|$ work, meaning the only solutions are $a = \pm b$.

$f(x + b) = f(x) + a$ means that, f is a bijection from the set of all numbers $x_b \bmod b$ to the set of all numbers $x_a \bmod a$ for a certain x_b, x_a .

Case 1: $|a| > |b|$

By Pigeonhole, it is impossible for f to be surjective since we can't get all $|a|$ residues modulo a from our $|b|$ residues modulo b .

Case 2: $|a| < |b|$

By Pigeonhole, it is impossible for f to be injective since some of the $|b|$ residues modulo b will overlap.

37 clubs; 10 hours.

Problem 14 (JMO 2021/1)

We claim the only solution is $f \equiv 1$, which obviously works.

We prove this using strong induction on n ,

Base Case: $n = 1$, $f(1) = f(1)^2$. Therefore, since $f(1) > 0$, $f(1) = 1$.

Inductive Step: $n = 1 \dots 2k$ to $n = 2k + 1$ and $n = 2k + 2$.

For $n = 2k + 1$, note that $(2k + 1)^2 + (2k^2 + 2k)^2 = (2k^2 + 2k + 1)^2 = (k^2 + (k + 1)^2)^2$.

$$1 = (f(k)f(k + 1))^2 = (f(k^2 + (k + 1)^2))^2 = f((k^2 + (k + 1)^2)^2) = f(2k + 1)f(2k^2 + 2k)$$

Therefore, since f takes natural numbers, $f(2k + 1) = 1$.

For $n = 2k$, note that $(2k)^2 + (k^2 - 1)^2 = (k^2 + 1)^2$.

$$1 = (f(k)f(1))^2 = f(k^2 + 1)^2 = f((k^2 + 1)^2) = f(2k)f(k^2 - 1)$$

Therefore, since f takes natural numbers, $f(2k) = 1$.

Conclusion: We are done by induction.

Problem 19 (USAMO 2002/4)

We claim that the only solution is $f(x) = kx$, which obviously works.

Claim: $f(0) = 0$.

Proof: Plug in $x = y = 0$.

Claim: f is odd.

Proof: Plug in $x = 0$ to get $f(x^2) = x f(x)$, plug $y = 0$ to get $f(-y^2) = -y f(y)$.

$$f(x^2 + 2x + 1) = f(x^2) + f(x + 1)$$

Therefore, we plug in $f(x^2) = x f(x)$ to get,

$$\begin{aligned} f(x^2 - y^2) &= x f(x) - y f(y) \\ \Leftrightarrow f(x^2 - y^2) &= x f(x) - y f(y) \\ \Leftrightarrow f(x^2 - y^2) &= f(x^2) - f(y^2) \end{aligned}$$

Since f is odd, for all $x \geq 0$ and $y \leq 0$,

$$\Leftrightarrow f(x + y) = f(x) + f(y)$$

However, since f is odd, this holds for all real x, y .

We plug in $t = x + 1$ into $f(x^2) = x f(x)$,

$$\begin{aligned} f((x + 1)^2) &= (x + 1)f(x + 1) \\ \Leftrightarrow f(x^2) + 2f(x) + f(1) &= (x + 1)f(x) + (x + 1)f(1) \\ \Leftrightarrow (x + 2)f(x) &= (x + 1)f(x) + x f(1) \\ \Leftrightarrow f(x) &= x f(1) \end{aligned}$$

This is as desired.

Problem 20 (Taiwan TST 2014)

We claim the only solution is $f(x) = x^3 - 1$, which obviously works.

Claim: $f(0) = -1, f(1) = 0$.

Proof: Plugging in $m = n = 0$ and $m = n = 1$ respectively, we get $f(0), f(1) \in \{-1, 0\}$. Since f is strictly increasing, we're done.

We let $g(x) = f(x) + 1$,

$$\begin{aligned} f(mn) &= f(m) + f(n) + f(m)f(n) \\ \Leftrightarrow g(mn) - 1 &= g(m) + g(n) + (g(m) - 1)(g(n) - 1) - 2 \\ \Leftrightarrow g(mn) &= g(m)g(n) \end{aligned}$$

We let $g(x) = e^{h(\log(x))}$.

$$\begin{aligned} \Leftrightarrow e^{h(\log(m)+\log(n))} &= e^{h(\log(m))} e^{h(\log(n))} \\ \Leftrightarrow h(\log(m) + \log(n)) &= h(\log(m)) + h(\log(n)) \end{aligned}$$

Since h is still strictly increasing, we use additive functions to get $h(x) = kx$. Therefore,

$$\begin{aligned} \Leftrightarrow g(x) &= e^{k \log(x)} = x^k \\ \Leftrightarrow f(x) &= x^k - 1 \end{aligned}$$

Since $f(2) = 7$, we have $k = 3$ and $f(x) = x^3 - 1$.

Problem 23 (IMO 2022/2)

We claim the only answer is $f(x) = \frac{1}{x}$, which works since $\frac{x}{y} + \frac{y}{x} \leq 2$ is only achievable when $y = x$.

From above, we get that for all solutions, we have $f(x) \leq \frac{1}{x}$. We call a number y to be *the* (only) friend of a number x if $x f(y) + y f(x) \leq 2$. However, plugging in $x = y$,

$$x f(x) + x f(x) \leq x \cdot \frac{1}{x} + x \cdot \frac{1}{x} = 2$$

Therefore, the only friend of each number is itself, meaning $x f(y) + y f(x) > 2$ for all $x \neq y$. We now plug in $y = x + t$ to get,

$$2 < x f(x+t) + (x+t) f(x) \leq \frac{x+t}{x} + \frac{x}{x+t} = 2 + \frac{t^2}{x^2 + xt}$$

As t approaches one, we must get that the middle $\leq$ will need to be an equality. Therefore, $f(x) \geq \frac{1}{x}$ as well, so we only have one solution.

57 clubs; 13 hours.